



**SYDNEY BOYS HIGH SCHOOL**  
**MOORE PARK, SURRY HILLS**

**2009**

**YEAR 11 ASSESSMENT TASK**  
**#1**

# Mathematics

## General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each question is to be returned in a separate booklet.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise stated.
- Full marks may not be awarded for careless or badly arranged work.

## Total Marks – 80

- Attempt questions 1 – 5
- All questions are of equal value.

Examiner: Ms F Nesbitt

This is an assessment task only and does not necessarily reflect  
the content or format of the Higher School Certificate

**Question 1 (16 marks)**

(a) Solve  $|x - 2| < 3$  and graph your answer on a number line 2

(b) Rationalise the denominator and simplify 2

$$\frac{3 + 2\sqrt{5}}{2\sqrt{5} - 1}$$

(c) Differentiate and simplify: 5

(i)  $\frac{x^2}{2} - 3\sqrt{x}$

(ii)  $\frac{6x + 5}{1 - 3x}$

(iii)  $\sqrt{2x + 8}$

(d) Solve the following equation.  $\log_2(x + 4) - \log_2(x - 2) = 1$  2

(e) A parabola has equation  $(x - 3)^2 = 8y$  2

Find: (i) the coordinates of its vertex

(ii) the equation of its axis of symmetry

(f) Find all the values of  $m$  for which the following quadratic equation 3

has real roots:

$$mx^2 - 8x + m = 0.$$

**Question 2 (16 marks)****Start a new booklet**

- (a) The function  $f(x)$  is defined by the rule: **3**

$$\begin{aligned} f(x) &= 0 \text{ if } x \leq 0 \\ f(x) &= 2x \text{ if } x > 0 \end{aligned}$$

- (i) Sketch the function in the Domain  $-2 \leq x \leq 2$

- (ii) Find the area between  $f(x)$  and the  $x$  axis.

- (b) For the function whose derivative is **2**

$$\frac{dy}{dx} = x^2(3x-1)(x-2),$$

determine the nature of the turning point at the point where  $x=0$

- (c) Solve:  $3^{x-5} = 7$  correct to 2 decimal places. **2**

- (d) Given that there is a root of  $kx^2 - 20x + k = 0$  at  $x = 3$ , **3**  
find the value of the other root.

- (e) Find, from first principles, the derivative of  $x^2 - 3$ . **3**

- (f) On a diagram, mark clearly the region for which **3**

$$y \geq -\sqrt{1-x^2}, y \geq -x \text{ and } y \leq 0 \text{ are true simultaneously.}$$

**Question 3 (16 marks)****Start a new booklet**

(a) Find  $\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 1}$  2

(b) If the roots of the equation  $x^2 - 5x + 2 = 0$  are  $\alpha$  and  $\beta$ , 4

Find the values of

(i)  $\alpha + \beta$

(ii)  $\alpha\beta$

(iii)  $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

(c) Solve: 5

(i)  $x^2 - 8x + 10 = 0$

(ii)  $9^x - 4(3)^x + 3 = 0$

(d) Find the domain over which the graph of the function 3

$y = \frac{2}{3}x^3 - \frac{5}{2}x^2 - 3x$  is concave up

(e) The probability that a train will be late on any given day is  $\frac{1}{5}$ . 2

Find the probability that, over a 3 day period, the train will be:

(i) late each day

(ii) late at least once.

**Question 4 (16 marks)****Start a new booklet**

- (a) The graph of  $y = f(x)$  passes through the point  $(3,1)$  and **3**

$$\frac{dy}{dx} = 1 + \frac{3}{x^2}. \text{ Find } f(x)$$

- (b) For the curve  $y = x^3 - 3x^2$  **9**

- (i) Find any stationary point(s)
- (ii) Determine the nature of the stationary point(s)
- (iii) Find any point(s) of inflexion
- (iv) Sketch the curve in the domain  $-1 \leq x \leq 3$  showing all the above features.

- (c) A parabola has equation  $x = 7 + 6y - y^2$  **4**

- Find
- (i) the coordinates of its vertex,
  - (ii) its focal length,
  - (iii) the equation of its directrix.

**Question 5 (16 marks)****Start a new booklet**

(a) Solve  $12 \times 8^{x-2} = \frac{3}{4^x}$  **2**

(b) Differentiate  $\frac{1}{\sqrt{x-1} - \sqrt{x}}$  **3**

You are not required to rationalise the denominator in your answer.

(c) If  $\alpha$  and  $\beta$  are the roots of  $x^2 - 8x + 5 = 0$ , **3**

find a quadratic equation with roots  $\alpha^2$  and  $\beta^2$

(d) Find the equation of the tangent and normal to the curve **4**

$y = x^3 - x^2$  at the point  $(2, 4)$

(e) The cost of running a long distance truck is **4**

$\left(\frac{1}{3}v^2 + 200\right)$  dollars per hour where  $v$  is the speed in km/h.

(i) Show that the cost for  $k$  kilometers is

$\frac{k}{v} \left(\frac{1}{3}v^2 + 200\right)$  dollars.

(ii) Find the value of  $v$  which will minimise the cost.

**End of the Paper**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right) \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x, \quad x > 0$

Q1 (a)  $|x-2| < 3$

$$\therefore -3 < x-2 < 3$$

$$\therefore -1 < x < 5$$



2

(b)  $\frac{3+2\sqrt{5}}{2\sqrt{5}-1} \times \frac{2\sqrt{5}+1}{2\sqrt{5}+1}$

$$= \frac{6\sqrt{5}+3+20+2\sqrt{5}}{20-1}$$

$$= \frac{23+8\sqrt{5}}{19}$$

2

(c) (i)  $\frac{d}{dx} \left( \frac{x^2}{2} - 3\sqrt{x} \right)$

$$= x - \frac{3}{2} x^{-\frac{1}{2}}$$

$$= x - \frac{3}{2\sqrt{x}}$$

1

(ii)  $\frac{d}{dx} \left( \frac{6x+5}{1-3x} \right)$

$$= \frac{(1-3x) \cdot 6 - (6x+5) \cdot (-3)}{(1-3x)^2}$$

$$= \frac{6 - 18x + 18x + 15}{(1-3x)^2}$$

$$= \frac{21}{(1-3x)^2}$$

2

(iii)  $\frac{d}{dx} (\sqrt{2x+8})$

$$= \frac{1}{2} (2x+8)^{-\frac{1}{2}} \cdot 2$$

$$= \frac{1}{\sqrt{2x+8}}$$

2

(d)  $\log_2(x+4) - \log_2(x-2) = 1$

$$\log_2 \frac{x+4}{x-2} = 1$$

$$\therefore \frac{x+4}{x-2} = 2$$

$$\therefore x+4 = 2x-4$$

$$x = 8$$

2

(e)  $(x-3)^2 = 8y = 4 \times 2y$

(i) Vertex is (3,0)

1

(ii) Axis of symmetry is  $x=3$

1

(f) For real roots

$$\Delta \geq 0$$

$$64 - 4 \times m \times m \geq 0$$

$$64 - 4m^2 \geq 0$$

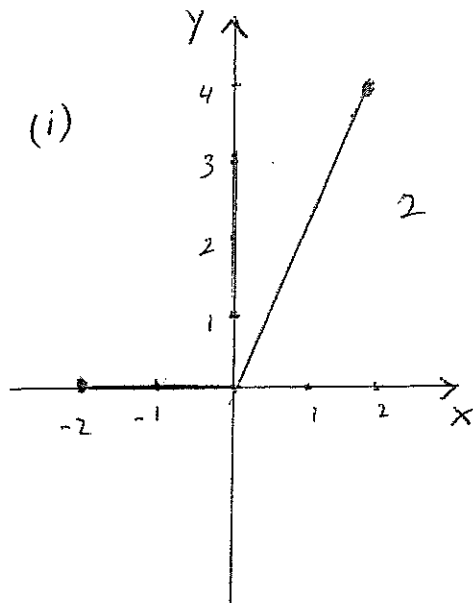
$$\therefore 4m^2 \leq 64$$

$$m^2 \leq 16$$

$$\therefore -4 \leq m \leq 4$$

3

2 (a) (i)



$$(ii) \text{ Area} = \frac{1}{2} \times 2 \times 4 = 4 \text{ u}^2$$

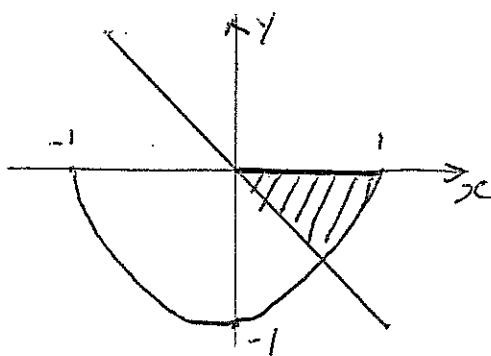
(b) at 0,  $\frac{dy}{dx} = 0$ ; it is positive at  $x = 0^-$  and  $x = 0^+$   
It is a horizontal point of inflexion.

(c)  $3^{x-5} = 7 \therefore x-5 = \frac{\ln 7}{\ln 3} \therefore x = 6.77$

(d) Product of roots =  $\frac{c}{a} = \frac{K}{K} = 1$ . Other root =  $\frac{1}{\lambda}$ .  
 $\therefore 3\lambda = 1 \therefore \lambda = \frac{1}{3}$

(e)  $f(x) = x^2 - 3$   
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$
$$= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 - 3) - (x^2 - 3)}{h}$$
$$= \lim_{h \rightarrow 0} \left( \frac{2xh + h^2}{h} \right)$$
$$= 2x$$

(f)



3

# YEAR 11 TASK 1 2009

## QUESTION 3

a)  $\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 1}$

$$= \lim_{x \rightarrow 1} \frac{x(x-1)}{(x-1)(x+1)}$$

$$= \lim_{x \rightarrow 1} \frac{x}{x+1} = \underline{\underline{\frac{1}{2}}}$$

b)  $x^2 - 5x + 2 = 0$

$$(x - \alpha)(x - \beta) = 0$$

i)  $\alpha + \beta = -b/a = \underline{5}$

ii)  $\alpha\beta = c/a = \underline{2}$

iii)  $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{5^2 - 2(2)}{2}$$

$$= \underline{\underline{\frac{21}{2}}}$$

c) (i)  $x^2 - 8x + 10 = 0$

$$x^2 - 8x + 10 + 6 = 6$$

$$(x - 4)^2 = 6$$

$$\underline{\underline{x = 4 \pm \sqrt{6}}}$$

(ii)  $9^x - 4(3^x) + 3 = 0$

$$(3^x)^2 - 4(3^x) + 3 = 0$$

$$(3^x - 3)(3^x - 1) = 0$$

$$3^x = 3 \text{ or } 3^x = 1$$

$$\underline{\underline{x = 1 \text{ or } x = 0}}$$

d)  $y = \frac{2}{3}x^3 - \frac{5}{2}x^2 - 3x$

is concave up when  $y'' > 0$

$$y' = 2x^2 - 5x - 3$$

$$y'' = 4x - 5$$

where  $y'' > 0$

$$4x - 5 > 0$$

$$x > 5/4$$

concave up

domain is  $x \in \mathbb{R} : x > 5/4$

e)  $P(\text{late}) = \frac{1}{5}$

$P(\text{late and late and late})$

$$= \frac{1}{5} \times \frac{1}{5} \times \frac{1}{5} =$$

$$= \underline{\underline{1/125}}$$

$P(\text{late at least once})$

$$= 1 - P(\text{not late} \times 3)$$

$$= 1 - (4/5)^3$$

$$= \underline{\underline{61/125}}$$

# Solutions to Q(4).

Q(4)

(a)  $\frac{dy}{dx} = 1 + 3x^{-2}$  (3)

$y = x - 3x^{-1} + c$  2

When  $x=3, y=1$

$1 = 3 - \frac{3}{3} + c$  1

$\therefore 1 = 2 + c \Rightarrow c = -1$

$\therefore y = x - \frac{3}{x} - 1$

(b)  $y = x^3 - 3x^2$

(i)  $\frac{dy}{dx} = 3x^2 - 6x$   
 $= 3x(x-2)$

(ii)  $\frac{dy}{dx} = 0$ , When  $x=0$ ,

$(0,0) (2,-4)$  3.

$f''(x) = 6x - 6$  2.

$f''(0) = -6 < 0$ , max

$f''(2) = 6 > 0$ , min

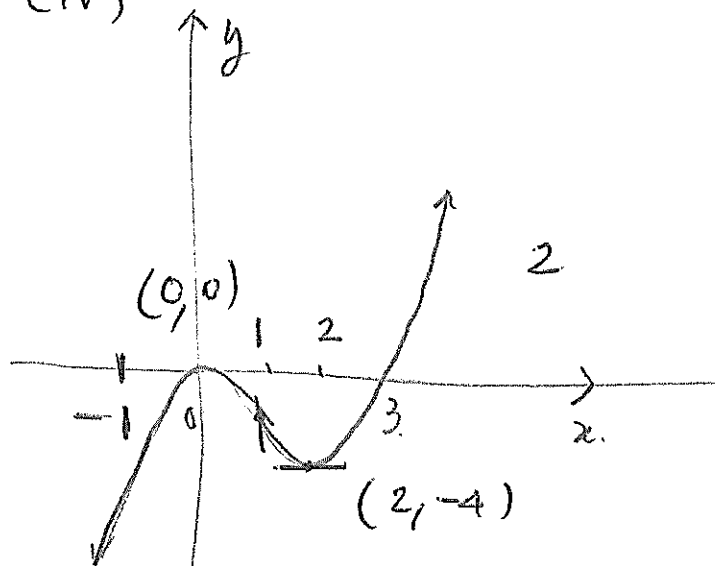
(iii)  $f''(x) = 0$  (9)

$x=1, y=-2$

| x        | $\frac{dy}{dx}$ | 1 | 2 |
|----------|-----------------|---|---|
| $f''(x)$ | -3              | 0 | 6 |

$\therefore (1, -2)$  is a pt of inflexion.

(iv)



$-y^2 + 6y + 7 = x$

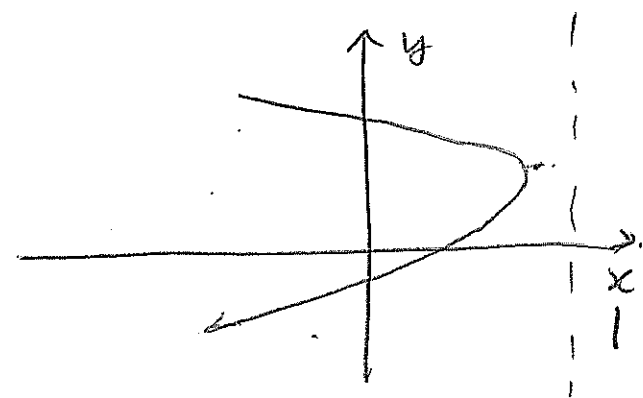
$y^2 - 6y - 7 = -x$  (4)

$(y-3)^2 - 16 = -x$

$(y-3)^2 = -(x-16)$  2

$\therefore$  vertex  $(16, 3)$

$a = \frac{1}{4}$  1



$x = 16\frac{1}{4}$

$x = \frac{65}{4}$

Q5.  
(a)  $3 \times 2^2 \times 2^{3x-6} = 3 \times 2^{-2x}.$

$$2^{3x-4} = 2^{-2x}.$$

$$3x-4 = -2x$$

$$5x = 4$$

$$x = \frac{4}{5}.$$

(b)  $y = \frac{1}{\sqrt{x-1} - \sqrt{x}} \times \frac{\sqrt{x-1} + \sqrt{x}}{\sqrt{x-1} + \sqrt{x}}.$

$$= \frac{\sqrt{x-1} + \sqrt{x}}{x-1 - x}.$$

$$y = -\sqrt{x-1} - \sqrt{x}.$$

$$\frac{dy}{dx} = -\frac{1}{2}(x-1)^{-\frac{1}{2}} - \frac{1}{2}(x)^{-\frac{1}{2}}$$

$$= -\frac{1}{2\sqrt{x-1}} - \frac{1}{2\sqrt{x}}.$$

(C) Method 1.

$$X = x^2 \Rightarrow \sqrt{X} = x.$$

$$\sqrt{X}^2 - 8\sqrt{X} + 5 = 0.$$

$$X + 5 = 8\sqrt{X}.$$

$$(X + 5)^2 = 64X.$$

$$X^2 + 10X + 25 = 64X.$$

$$X^2 - 54X + 25 = 0.$$

Method 2.

$$\alpha + \beta = 8$$

$$\alpha\beta = 5.$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta.$$

$$\begin{aligned} &= 64 - 10 \\ &= 54 = -\frac{b}{a}. \end{aligned}$$

$$\begin{aligned} \alpha^2\beta^2 &= (\alpha\beta)^2 \\ &= 25 = \frac{c}{a}. \end{aligned}$$

$$\text{let } a = 1$$

$$x^2 - 54x + 25 = 0$$

$$(ii) \quad C = \frac{k}{3}v + 200kv^{-1}$$

$$\frac{dC}{dv} = \frac{k}{3} - 200kv^{-2}$$

$$\frac{k}{3} - 200kv^{-2} = 0.$$

$$v^2 = 600$$

$$v = \pm 10\sqrt{6}.$$

Taking the positive velocity.

$$v = 10\sqrt{6}$$

$$\frac{d^2C}{dv^2} = 400kv^{-3}.$$

$$\text{at } v = 10\sqrt{6}$$

$$\frac{d^2C}{dv^2} = \frac{400k}{(10\sqrt{6})^3} > 0.$$

minima.

$\therefore$  velocity that minimises cost is

$$\cancel{10\sqrt{6}} \quad 10\sqrt{6} \text{ km/hr.}$$